GRISHMA BALGI

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Education

New York University (NYU) **Aug. 2025 – May 2027**
M.Sc. in Mechatronics and Robotics **GPA: 3.834 / 4.0**
Coursework: Foundation and Mathematics of Robotics · Advanced Mechatronics · Real-Time Embedded Systems · Robot Perception

Veermata Jijabai Technological Institute (VJTI) **Aug. 2021 – Jun. 2025**
B.Tech. in Mechanical Engineering, Honors in Automobile Engineering **GPA: 9.04 / 10.0**

Technical Skills

Programming: Python, C/C++, MATLAB/Simulink, NumPy, SciPy, Pandas, Matplotlib
Robotics & Modeling: MuJoCo, forward/inverse kinematics, Jacobians, passive dynamics, reduced-order modeling
Sensing & Control: Dynamixel, Arduino, STM32/Mbed OS, HX711 load cells, IMUs, I2C, serial communication
Design & Experiments: SolidWorks, Creo, FDM fabrication, test-rig construction, force and motor-position logging
Data & Interfaces: scikit-learn, signal filtering, dynamic time warping, Tkinter, PyUSB

Research Experience

Prema Lab, New York University **Sep. 2025 – Present**
Graduate Researcher | Brooklyn, NY

Underwater Soft Robotics – Sim-to-Real Modeling | Python, MuJoCo

- Built a **CAD-derived MuJoCo model of an 18-segment TPU 95A tentacle**, representing compliance through **17 passive hinge joints and two tendons**; parameterized joint stiffness and damping to simulate deflection, spring-back and settling.
- Implemented **buoyancy and hydrodynamic-drag force variants** to investigate underwater passive dynamics; developing a **sim-to-real calibration workflow** to compare simulated motion with physical experiments and identify model discrepancies.

Automated Fin Fabrication & Instrumented Testing | Python, Dynamixel, Arduino

- Fabricated an **instrumented fin-testing rig** and automated Silhouette Portrait 4 fabrication through Python/PyUSB; converted fin geometries into coordinate-based cutting paths using piecewise-linear curve approximation.
- Built a **Tkinter experiment-control interface** integrating Dynamixel actuation, Arduino/HX711 force acquisition and serial clamp commands; exposed waveform, amplitude and frequency settings with **four-second averaged load-cell tare**.
- Logged **force, commanded position and measured motor position** across fin geometries and actuation settings, creating trial records for comparative performance analysis and ongoing refinement of the measurement protocol.

Industry Experience

Schneider Electric **Jun. 2024 – Jun. 2025**
Engineering Intern | Mumbai, India

- Developed **reduced-order models for digital-twin development** using Creo, Altair romAI and Twin Activate; trained a bracket-response surrogate on FEA-derived data to predict **six stress and two load outputs from two strain inputs**.
- Built vibration-acquisition components for **circuit-breaker condition monitoring**; evaluated sensor range, sensitivity and size, and developed Python preprocessing and classification workflows for **four spring-removal conditions**.
- In related final-year work, compared **five classifiers** using an **80:20 train-test split**; Random Forest and XGBoost reached **99.98% accuracy on synthetic test data**, with **two errors in 12,000 test samples**.

Projects

UR10e Kinematics & Velocity Control
Python | NumPy | SciPy | MuJoCo

- Implemented **Denavit-Hartenberg forward kinematics and numerical inverse kinematics** for a UR10e manipulator in a supplied MuJoCo framework, mapping joint configurations to end-effector poses and solving joint angles for Cartesian targets.
- Computed **Jacobian-based velocity mappings and pseudoinverse joint-velocity solutions** to translate task-space commands into joint motion; simulated the resulting manipulator trajectories in MuJoCo.

KinetiKey – Embedded Gesture Authentication
C++ | STM32L475 | Mbed OS | IMU

- Implemented an **STM32 gesture-authentication pipeline** combining **400 kHz I2C** IMU acquisition, low-pass filtering and dynamic time warping to recognize an enrolled **three-gesture sequence**.
- Bounded DTW computation with a **30-sample search band, two-row workspace and 200-sample gesture buffers**; implemented interrupt-driven enrollment, LED state feedback and three-attempt lockout.

Vision-Based Robot Navigation
Python | Visual Place Recognition | Topological Graphs | Dijkstra

- Implemented **visual place recognition, topological graph construction and Dijkstra route planning** within a supplied navigation framework; added return-route connections and arrival verification for camera-only, image-goal navigation.
- Earned **10/10 points with SPL 1.0** in a saved course-server evaluation: **718 steps, 3.13 cm arrival-position error and 0.53 degree heading error**; preserved the evaluated implementation for repeatable comparisons.